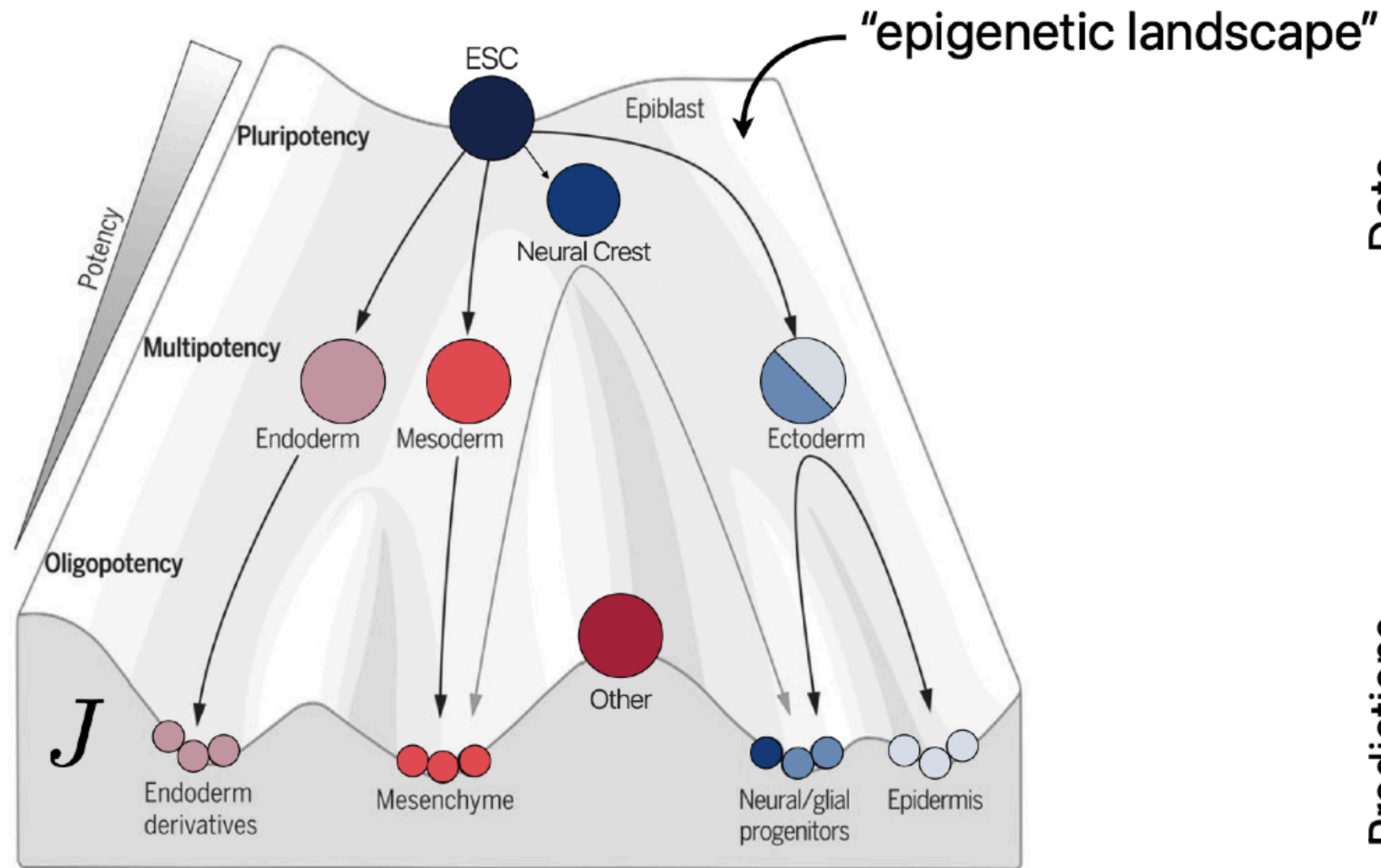
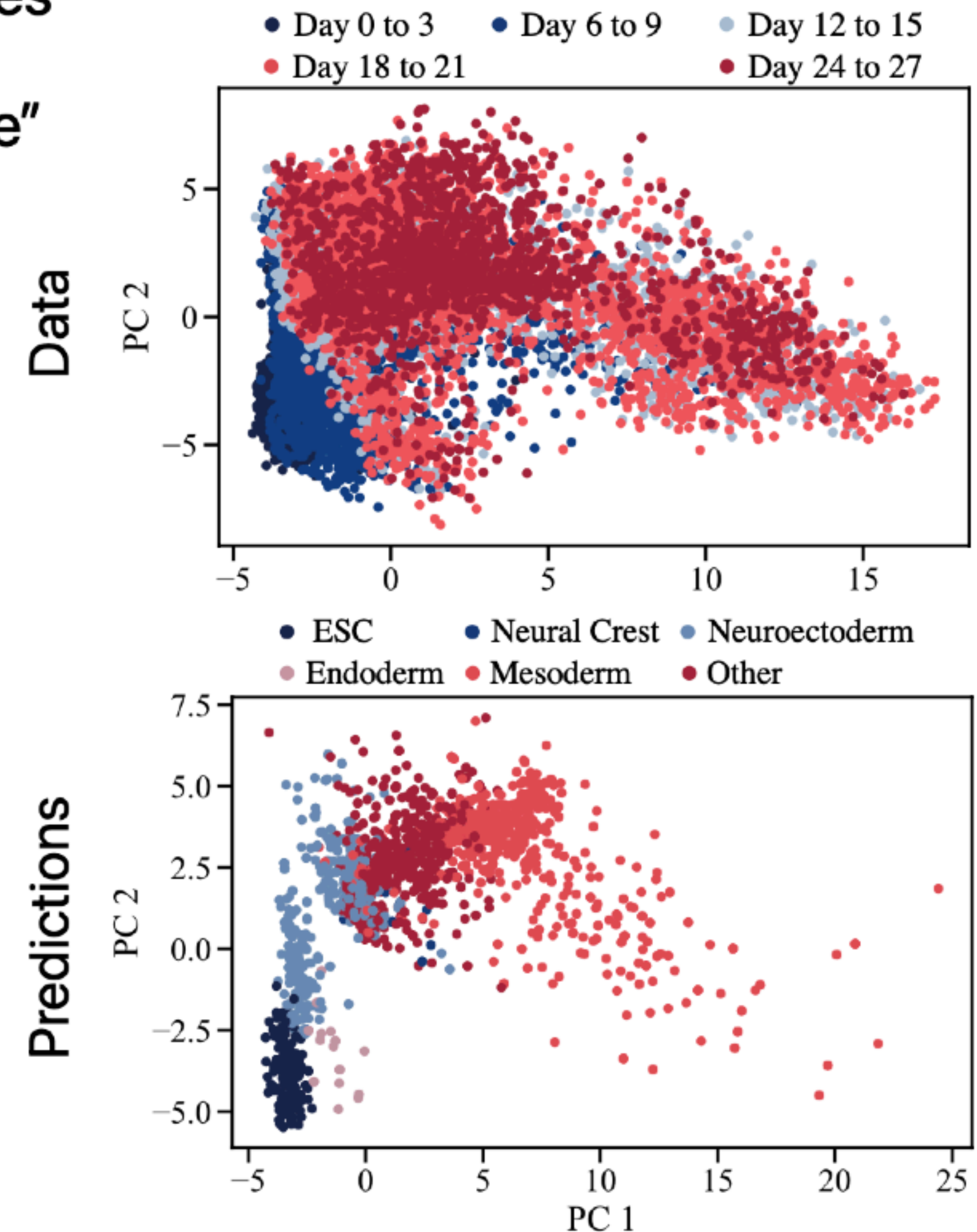


Application: Modeling Cell Trajectories

Differentiation of embryonic stem cells into diverse cell lineages



[Bunne+22, Moon+19, Zalc+21, Tong+21, Hashimoto+16]



Takeaways & Preview

- Dynamic OT = geodesic view of transport, links geometry, PDEs, and ML.
- Benamou–Brenier = kinetic energy minimization
- Continuity equation ensures mass conservation
- McCann interpolation = explicit geodesic formula
- Next lecture:
 - Gradient Flows
 - Entropic/stochastic variants (Schrödinger bridges)